

Charles Bradley

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EDUCATION

California Polytechnic State University

San Luis Obispo, CA

Bachelor of Science in Mechanical Engineering - Mechatronics Concentration

Expected May 2028

- **GPA (Major):** 3.52 / 4.0
- **Relevant Coursework:** Mechanics of Materials, Engineering Dynamics, Engineering Statics, Thermodynamics, Manufacturing Processes: Material Removal and Metal Casting
- **Activities & Societies:** Professional Project Chairman, Sigma Phi Delta Engineering Fraternity

PROJECTS

Low-Cost GNSS Tide Gauge Research | *California Polytechnic State University*

Sept 2026 – Present

- Developing a MATLAB script to calculate calibration constants for an uncalibrated sensor using synchronized data points from a calibrated reference gauge
- Installed and deployed radar-based water level gauges in Morro Bay to support coastal water-level monitoring
- Contributing to the development of low-cost tide gauges designed to expand sea-level monitoring among currently unmonitored locations

3D-Printed Fixed Wing Aircraft | *Sigma Phi Delta*

Sept 2026 – Present

- Leading a 12-member fraternity engineering team designing, manufacturing, and testing a fully 3D-printed fixed-wing aircraft for the CSU 3D-Printed Fixed-Wing Aircraft Competition
- Developing ANSYS and MATLAB flight simulations to evaluate aerodynamics, estimate flight duration, and optimize trajectories across wing and fuselage design iterations
- Translating competition requirements into design constraints to optimize aircraft weight, structural integrity, manufacturability, and flight performance

Tactical Autonomous Robotic Sentinel | *Sigma Phi Delta*

Jan 2026 – Present

- Leading development of a proof-of-concept autonomous robot using servo-actuated joints
- Implementing voice-command recognition on an ESP32 to decode simple commands and map them to predetermined servo positions
- Designing a scalable mechatronics platform for future expansion into vision-based automation and artificial intelligence

Air Motor | *California Polytechnic State University*

Nov 2025 – Dec 2025

- Collaborated on a five-person team to machine a functional compressed-air motor
- Manufactured precision components using five-axis CNC mills and manual lathes to meet dimensional tolerances within ± 0.005 in
- Assembled and tested the motor under controlled conditions, achieving speeds of approximately 2900 RPM

Vacuum Chamber | *Milwaukee Tool*

Jan 2023 – Apr 2023

- Designed a 90 ft³ vacuum chamber to enable in-house high-altitude testing
- Calculated pump sizing and chamber pressure requirements to reach 12 PSIA with an evacuation time of under 60 seconds.
- Developed proof-of-concept and detailed CAD models of the chamber structures, supported by strength and deflection analysis
- Achieved a vacuum capable of simulating altitudes up to 29,500 ft

CERTIFICATIONS

Certified SolidWorks Associate (CSWA) | *Dassault Systèmes*

Autodesk Inventor Certified User | *Certiport*

Lean Six Sigma Fundamentals | *LinkedIn*

TECHNICAL SKILLS

Software: SolidWorks, Autodesk Inventor, MATLAB, Siemens NX

Hardware: Hand Drafting, GD&T, Soldering, Machining (Manual Lathe & CNC Mill)